Vision-Based Precision Pose Estimation for Autonomous Formation Flying

Neha K S (PES1UG24CS581) and Mohith N (PES1UG24CS580)
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***Abstract*—**Tight formation flight requires the relative pose of a leader aircraft to be known far more precisely than GPS or inertial sensing allows at close separations. We reproduce a hybrid vision-based approach (Punnoose, Stanford CS229, 2020) in which an artificial neural network (ANN) classifies detected aircraft features from a single image into a coarse, discrete pose label, and a particle filter (PF) informed by that label refines it. Because the original work releases no data, we build a synthetic pipeline: a 14-feature aircraft model, a pinhole camera, normal-based self-occlusion, and a noisy feature detector. The ANN (42 → 100 → 100 → 4800, implemented from scratch in NumPy) reaches 69.6% top-1 and 96.0% top-5 accuracy over 4800 pose cells, with 98.3% of predictions within one neighbouring cell. Combined with a PF, it reduces mean position error from 5.32 m (ANN alone) to 0.62–0.96 m and roll error from 4.51° to 0.80–1.09°. We also implement the original paper's own future-work suggestion, a Levenberg–Marquardt minimisation of the reprojection error initialised from the ANN's top-3 classes, which achieves 0.25 m and 0.18° at low detector noise. Under increased pixel noise, the temporal Gaussian PF overtakes the per-frame refinement, motivating a fused approach. Absolute numbers are not comparable to the original paper because our simulator is cleaner.

***Keywords*—**pose estimation, formation flight, neural network classification, particle filter, reprojection error, bundle adjustment

I. INTRODUCTION

In autonomous formation flight, a follower aircraft F carries a camera that is gimballed to always point at a leader aircraft L. From a single image, the follower must estimate the leader's relative pose, namely its 3-D position and its roll about the line of sight, precisely enough to hold tight formation. GPS and inertial sensors are not accurate enough at these separations. A pure end-to-end regressor, on the other hand, provides no uncertainty estimate and ignores the dynamics of the problem. This motivates a hybrid design in which a learned model supplies a robust coarse estimate and a probabilistic filter exploits dynamics and geometry to refine it.

Our assigned problem is the reproduction of Punnoose [1], where an ANN classifies detected aircraft features into a discrete pose label and a particle filter, informed by that label, refines the estimate. Our contributions are:

1) A fully synthetic, reproducible data-generation pipeline (aircraft model, camera, occlusion, detector, trajectories), since the original work releases no data.

2) A from-scratch NumPy implementation of the ANN pose-label classifier and of three particle-filter configurations (two from the paper and one with a Gaussian likelihood and roughening).

3) An extension following the paper's own future-work suggestion: a single-view Levenberg–Marquardt refinement of the reprojection error, started from the ANN's top-3 classes.

4) An empirical comparison of all methods on 10 simulated approach trajectories, including robustness to detector pixel noise.

II. PROBLEM FORMULATION

The state to be estimated is a 4-DoF relative pose of the leader with respect to the follower: its 3-D position (x, y, z) and its roll about the line of sight. The camera is gimballed to point at the leader, so the observation is a set of 2-D image locations of detected aircraft features. The goal is to recover the 4-DoF pose from a single image (per-frame estimation) and, where available, from a sequence of images (temporal filtering).

III. DATASET

No data is released with the original paper, so we generate it synthetically.

A. Aircraft and camera model. The aircraft is modelled by 14 features (nose, canopy, wing and stabiliser tips, nozzles, intakes, and others) projected through a pinhole camera with a 1280 × 960 px image and focal length f = 1600 px. Each feature has an outward normal; features facing away from the camera are treated as occluded. This normal-based test is a stand-in for full ray tracing.

B. Detector model. The simulated detector adds 0.5 px Gaussian noise to each feature location and misses 2% of visible features. A sample is the 42-vector [u, v, occluded] × 14.

C. Labels. Each pose is assigned to one of 4800 cells, following Table I of the original paper: x has 10 bins over [−100, 50] m, y has 10 bins over [−50, 50] m, z has 8 bins over [−20, 20] m, and roll has 6 bins over [−45°, 45°]. The cell size is therefore 15 m × 10 m × 5 m × 15°.

D. Splits. We draw 300,000 training, 30,000 validation and 30,000 test poses uniformly in this box, with range at least 15 m.

E. Trajectories. For filter evaluation we generate 10 random 10 s approach trajectories sampled at 10 Hz. The follower flies from about 90 m to about 30 m behind the leader while the leader manoeuvres.

IV. METHODS

A. ANN pose-label classifier. The classifier is a fully connected network with architecture 42 → 100 → 100 → 4800, ReLU hidden activations, a softmax output, and the Adam optimiser, as in the original paper. Forward and backward passes are written by hand in NumPy, with no deep-learning framework.

B. Particle filter. Particles are initialised from the ANN label plus an error pool, defined as the (true − label) differences measured on validation data. At each step, particles are

propagated through the relative dynamics, in which the leader's unknown speed, pitch and roll appear as a uniform disturbance. Each particle's 14 features are projected and compared with the detected features, ignoring occluded ones, and particles are weighted by a likelihood of the reprojection discrepancy. Resampling draws floor(αN) particles from the weighted set and the remainder from the ANN label plus the error pool. We implement three configurations (Table I).

TABLE I. Particle-filter configurations.

Config.	Likelihood	N	α	Notes
Paper 1	Inverse MSE	1000	0.9	Every 10 cycles
Paper 2	Inverse MSE	5000	0.9	Every cycle
Gaussian	Gaussian, roughening	2000	0.5	Our variant

C. Extension: reprojection-error refinement. Following the original paper's suggestion for future work, we perform a Levenberg–Marquardt minimisation of the reprojection error over the 4-DoF pose. The optimisation is started from each of the ANN's top-3 classes, and the lowest-cost solution is kept. This is a single-view form of bundle adjustment. In the results it is labelled "Gauss-Newton refinement", which refers to the Gauss-Newton-type least-squares step within the Levenberg–Marquardt scheme.

V. IMPLEMENTATION

The code is written in Python using only NumPy, SciPy and Matplotlib. The package `poseest/` contains modules for the aircraft model, projection, occlusion and detector, label discretisation, dataset generation, the ANN, dynamics, the particle filter, the refinement and the pipeline. Scripts cover ANN training (about 9 minutes on one CPU core), the main experiment, the noise sweep, and a live demo with GIF output. Unit tests cover label round-trips, projection scaling, occlusion, the refinement, and trajectory validity.

VI. RESULTS

A. Classifier accuracy. On the test set the ANN reaches 69.6% top-1 accuracy (the paper reports 54%), 96.0% top-5 accuracy, and 98.3% of predictions fall within one neighbouring cell. The mean cell-centre error is 5.6 m in position and 4.3° in roll. Almost all of the remaining error is therefore due to discretisation, as the original paper also observed.

B. Trajectory estimation. Table II reports the mean ± standard deviation over the 10 trajectories.

TABLE II. Pose-estimation error over 10 simulated approach trajectories.

Method	Position error (m)	Roll error (deg)
ANN only (cell centre)	5.32 ± 0.31	4.51 ± 0.58
ANN + PF, paper cfg 1 (N=1000)	0.96 ± 0.25	1.09 ± 0.20
ANN + PF, paper cfg 2 (N=5000)	0.70 ± 0.09	0.80 ± 0.14
ANN + PF, Gaussian (N=2000)	0.62 ± 0.09	0.96 ± 0.09
ANN + refinement (ours)	**0.25 ± 0.04**	**0.18 ± 0.03**

Fig. 1 (omitted in this version; see the original write-up). Mean position and roll error over time on a log scale, and mean position error versus detector pixel-noise standard deviation (ANN not retrained).

C. Robustness to detector noise. The refinement is clearly best at low noise. Because it operates on each frame independently, however, its error grows with detector noise, and beyond roughly 2 px the temporal Gaussian PF overtakes it. The ANN-only baseline is almost flat in noise but remains limited by cell size.

VII. DISCUSSION

The ANN provides a reliable coarse label (about 70% exact, 98% within one cell), but its precision is capped by the 15 m × 10 m × 5 m × 15° cell size. In the original paper, the PF changed the estimate little. In our experiments the ANN-informed PF cuts the error about 7× (5.3 m to 0.6–0.7 m). We attribute this to our cleaner simulator: an exact camera model, Gaussian noise, no outliers, and a 4-DoF state. Absolute numbers are therefore not comparable to those of the original paper.

The reprojection-based refinement gives the lowest error at low noise (0.25 m, 0.18°), supporting the paper's suggestion that the ANN is best used as a robust initialiser. The refinement is per-frame and does not exploit temporal information, which explains why the PF is preferable at higher noise. A natural next step is to fuse the two, using the refinement as the measurement in the PF.

VIII. LIMITATIONS

Data is synthetic only. Occlusion and dynamics are simplified (normal-based occlusion rather than ray tracing). The CNN feature detector is assumed, not built; detections are simulated with Gaussian noise and random misses. The noise sweep does not retrain the ANN.

IX. CONCLUSION

We reproduced a hybrid ANN and particle-filter pipeline for vision-based relative pose estimation in formation flight using a fully synthetic data pipeline. The ANN provides robust coarse labels, the PF reduces error by roughly a factor of seven, and an added Levenberg–Marquardt reprojection refinement reaches 0.25 m and 0.18° at low noise. Fusing the refinement with the PF is the most promising direction for combining low-noise precision with robustness at high noise.

REFERENCES

[1] Punnoose, Stanford CS229 course project, 2020. *(Details taken from the project write-up only; please verify the full citation against the original.)*